

AdaBoost

AdaBoost combines multiple weak learners (typically simple models like decision stumps) to create a strong classifier. Here's a brief outline of the procedure:

1. Initialize Weights:

- Start with equal weights for all training instances.
- If there are N data points, each point initially has a weight of $\frac{1}{N}$.

2. Train Weak Learners:

- For each round t :

1. **Train the Weak Learner:** Train a weak learner h_t on the weighted training data.

2. **Calculate the Error Rate:** Calculate the weighted error rate ϵ_t of the weak learner:

$$\epsilon_t = \frac{\sum_{i=1}^N w_i \mathbb{I}(h_t(x_i) \neq y_i)}{\sum_{i=1}^N w_i}$$

where $\mathbb{I}$ is the indicator function, $h_t(x_i)$ is the prediction of the weak learner on instance x_i , and y_i is the true label.

3. **Compute the Weight α_t of the Weak Learner:**

$$\alpha_t = \frac{1}{2} \ln \left(\frac{1 - \epsilon_t}{\epsilon_t} \right)$$

- The $\ln$ function helps in distinguishing between the performance of the weak learners effectively.
- When the error ϵ_t is small, the weight α_t becomes large, increasing the influence of a highly accurate weak learner.
- Conversely, when ϵ_t approaches 0.5 (random guessing), α_t becomes smaller, reducing the influence of a weak learner whose performance isn't significantly better than random guessing.

4. **Update the Weights of the Training Data:**

- Increase the weights of the misclassified instances and decrease the weights of the correctly classified instances:

$$w_i \leftarrow w_i \cdot \exp(\alpha_t \cdot \mathbb{I}(h_t(x_i) \neq y_i))$$

- The exponential function magnifies the effect of the weight adjustment. For misclassified instances, it increases the weight multiplicatively by a factor of $\exp(\alpha_t)$.
- Correctly classified instances see their weights decreased (since the exponent is zero for correct classifications).

5. **Normalize the Weights:** Normalize the weights so that they sum up to 1.

3. Final Prediction:

- The final model is a weighted sum of the weak learners' predictions:

$$H(x) = \text{sign} \left(\sum_{t=1}^T \alpha_t \cdot h_t(x) \right)$$

where T is the total number of weak learners.